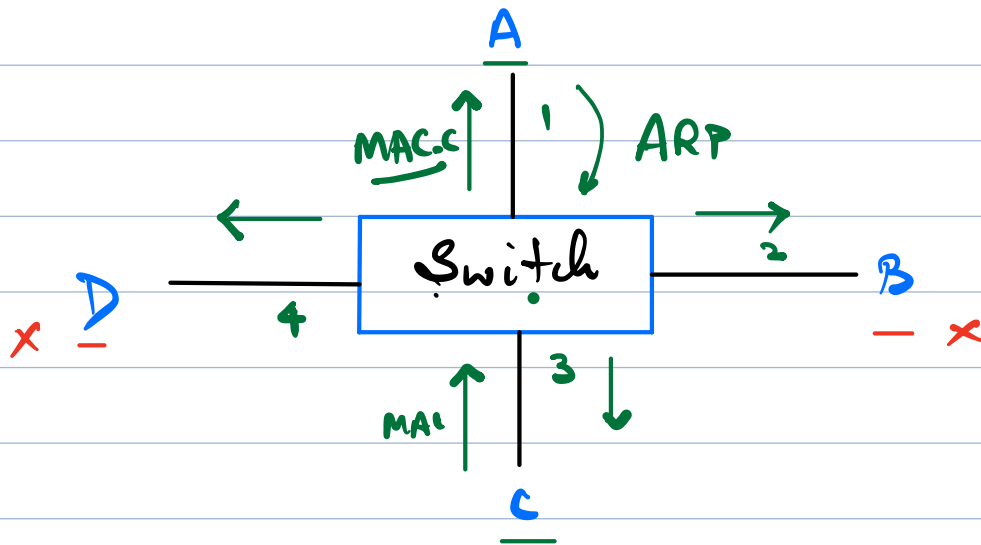




- 1) MAC
- 2) IP Address

ARP Table

IP → MAC Address

IP.C → MAC.C

MAC Address

MAC → Port

MAC.A → 1

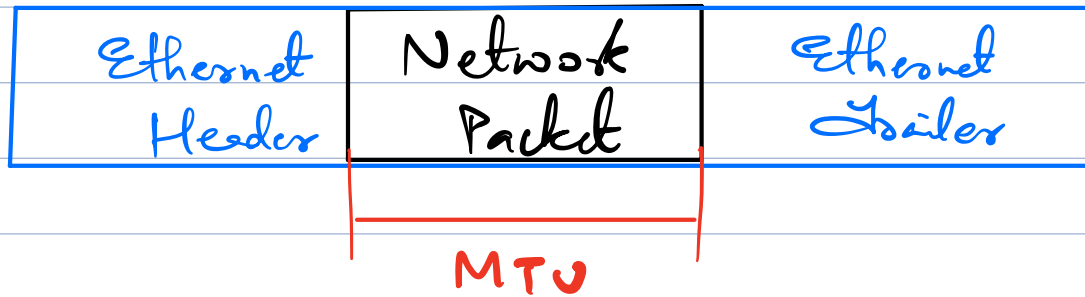
MAC.C → 3

ARP ⇒ Address Resolution Protocol.

A — Wifi — ISP Network — R1 → R2 — Google Net. — Google Server

IP Address

MTU (Max. Transmission Unit.)



Link Type

Ethernet
Wifi
Jumbo frames (Data Center)

MTU

1500 Bytes
1500 Bytes
Upto 9000 Bytes

Ethernet frame Structure



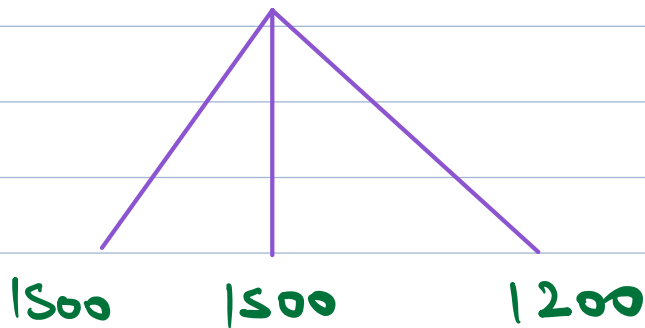
Preamble	Dest. MAC	Src. MAC	Ether Type	Payload	FCS
8 Bytes	6 Bytes	6 Bytes	2 Bytes	46 - 1500 Bytes	4 Bytes

IP PDU

IP Header 20 Bytes	TCP Segment 1480 Bytes
-----------------------	---------------------------

1500 Bytes

4200 Bytes.



} IP fragmentation

IP Addresses & Subnetting.

Global Uniqueness (for public IPs)
Hierarchical Structure

IPv4 Address format

32 bit number written as four decimal octets separated by dots:

8 bit no $\Rightarrow [0 - 255]$

8bits

192

.

8bits

168

.

8bits

1

.

8bits

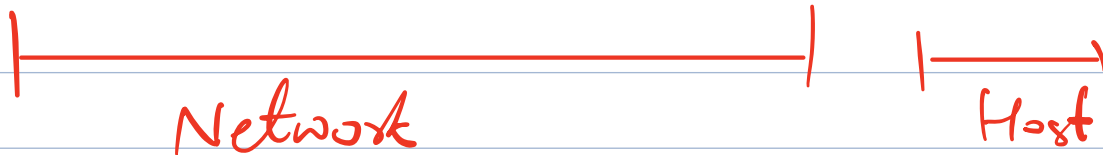
5

Total no. of IPv4 Addresses ?! = 2^{32}

$\approx 4.3 \text{ Billion}$

Network ID & Host ID

192 . 168 . 1 . 5



Subnet Mask $\Rightarrow$ 32 bit long no.

7 6 5 4 3 2 1 0
| 0 | 1 1 0 1 0 1
M = 0 0 1 0 0 0 0 0

Network

Host

192 . 168 . 1 . 5

11000000 10100000 00000001 00000101

8 1111111 1111111 1111111 00000000

11000000 10100000 00000001 00000000

198.168.1.0

8 Host Bits $\Rightarrow 2^8 - 2$

Broadcast $\Rightarrow$ All host bits are 1

198.168.1.255 $\rightarrow$ Broadcast

198.168.1.0 $\rightarrow$ Network

198.168.1.1 to 198.168.1.254

Class	First Octet	Network	Host	Mask
-------	-------------	---------	------	------

A	1 - 126	8	24	255.0.0.0
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B 128-191 16 16 255.255.0.0

C 192-223 24 8 255.255.255.0

CIDR Notation

/N prefix to indicate the no. of network bits

format: IP Address / Prefix-length

192.168.1.5 / 24

↳ first 24 bits are network bits

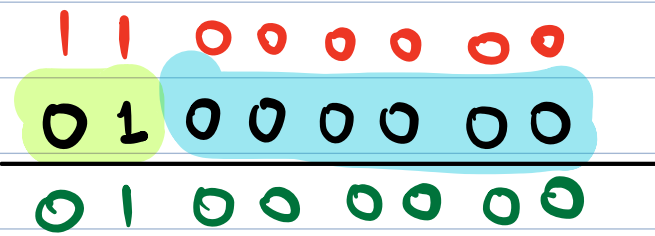
Q for 192.168.1.64 / 26

Network : 192.168.1.64

Hosts : : 192.168.1.65 - 192.168.1.126

Broadcast : 192.168.1.127

Usable : : $2^6 - 2 = 62$

64 ⇒ 

255. 255. 255. 192

BroadCast ⇒ 192.168.1. 01111111

Common CIDR Reference

CIDR	Subnet Mask	Usable Hosts	Common Use
/8	255.0.0.0	16,777,214	Large ISPs
/16	255.255.0.0	65,534	Large enterprises
/24	255.255.255.0	254	Home, office networks
/25	255.255.255.128	126	Medium subnets
/26	255.255.255.192	62	Small subnets
/27	255.255.255.224	30	Small offices
/28	255.255.255.240	14	Very small subnets
/30	255.255.255.252	2	Point-to-point links
/32	255.255.255.255	0 (single host)	Host routes

Public vs Private IP Addresses

Range	CIDR	Size	Common Use
10.0.0.0 – 10.255.255.255	10.0.0.0/8	16.7M addresses	Large corps, cloud VPCs
172.16.0.0 – 172.31.255.255	172.16.0.0/12	1.04M addresses	Medium networks
192.168.0.0 – 192.168.255.255	192.168.0.0/16	65,536 addresses	Home routers, offices

Loopback $\Rightarrow$ 127.0.0.1 / 8

Other Reserved Ranges

Address / Range	Purpose
0.0.0.0	Unspecified / "any" address
127.0.0.0/8	Loopback (entire /8 reserved)
169.254.0.0/16	APIPA — automatic self-assigned when DHCP fails
255.255.255.255	Limited broadcast (all hosts on local subnet)

DHCP: Dynamic Host Config. Protocol.

D : Discover

O : Offer

R : Request

A : Acknowledge.